23130

C23-EC-302

23130

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – THIRD SEMESTER EXAMINATION

ELECTRONIC CIRCUITS—I

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Draw the circuit diagram of half wave rectifier and its output waveform.

2. Write the expressions for RMS value and average value (DC value) of voltages and current of full wave bridge rectifier.

3. List the factors affecting the operating point.

4. Draw the h-model of CE transistor configuration.

5. Define the terms Gain in db and bandwidth of an amplifier.

6. Draw the practical single-stage transistor CE amplifier.

7. Classify power amplifiers based on period of conduction.

8. List the merits of negative feedback amplifiers.

9. List the advantages of crystal oscillators over other types of oscillators.

10. Compare positive and negative feedback.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Explain the working of full wave Bridge rectifier circuit with wave forms.

12. *(a)* Explain the working of a simple Zener regulator. 5

(b) Explain the concept of DC and AC load lines. 5

13. Explain the fixed bias circuit and list its drawbacks.

14. Explain the working of two-stage transformer coupled amplifier with circuit diagram and also draw its frequency response.

15. Explain the working of complementary symmetry Push-pull power amplifier with circuit diagram.

16. Draw the circuit diagram of double tuned amplifier and explain its operation with frequency response curve.

17. Draw the block diagrams of voltage series, current series, current shunt and voltage shunt feedback amplifiers and compare their characteristics.

18. Explain the working of Hartley oscillator with a circuit diagram.

★★★